

Commands used to generate logs

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2024-05-28

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1 Ignored logs

1.1 Ignored namespaces and users

The following namespaces are ignored:

- falco
- kube-flannel

The following users/SAs are ignored:

- system:serviceaccount:kube-flannel:flannel

1.2 Endpoints different than api/apis

We will ignore endpoints which are not the regular API ones.

- Verb: any
- Users: any
- Objects: any
- Endpoints: any other than /api or /apis

1.3 API Server watching objects

The API server monitors core components with a 10minute timeout. It remains unclear what the API server watches in other groups: we probably need to deploy more components to see what the API server watches.

- Verb: `watch`
- Users: `system:apiserver`
- Objects: `configmaps`, `clusterroles`, `namespaces`, `serviceaccounts`, `resourcequotas`, `clusterrolebindings`, `rolebindings`, `secrets`, `nodes`, `Pods`, `roles`
- Namespaces: none, apart from `configmaps` in `kube-system`. Additionally, if legacy service accounts are used, the API server also watches `kube-apiserver-legacy-service-account-token-tracking`, a CM in `kube-system`.

1.4 kube-controller-manager watching objects

The `kube-system` namespace watches objects, similar to the API server. However, `ConfigMaps` are watched over non-namespaced objects and also `certificateigningrequests` are watched.

I believe that the objects watched by the Kube Controller Manager are specific to what is deployed in the cluster. For example, if the cluster has a deployment, the Kube Controller Manager will watch also `deployments`.

- Verb: `watch`
- Users: `system:kube-controller-manager`
- Objects: same as the API server, plus `certificateigningrequests`
- Namespaces: none

1.5 kube-controller-manager getting and creating tokens for GC and RQ controllers

The `kube-controller-manager` gets every less than an hour the `serviceaccounts` `/api/v1/namespaces/kube-system` and `/api/v1/namespaces/kube-system/serviceaccounts/resourcequota-controller`. It then creates tokens for them, which expire in an hour.

1.5.1 1

- Verb: `get`
- Users: `system:kube-controller-manager`, groups `system:authenticated`
- Objects: `/api/v1/namespaces/kube-system/serviceaccounts/generic-garbage-collector`

1.5.2 2

- Verb: `get`
- Users: `system:kube-controller-manager`, groups `system:authenticated`
- Objects: `/api/v1/namespaces/kube-system/serviceaccounts/resourcequota-controller`

1.5.3 3

- Verb: `create`
- Users: `system:kube-controller-manager`, groups `system:authenticated`
- Objects: `/api/v1/namespaces/kube-system/serviceaccounts/generic-garbage-collector/token`

1.5.4 4

- Verb: `create`
- Users: `system:kube-controller-manager`, groups `system:authenticated`
- Objects: `/api/v1/namespaces/kube-system/serviceaccounts/resourcequota-controller/token`

1.6 Kube Scheduler watching objects

The Kube Scheduler watches pods, nodes, namespaces, always with a 10-minute timeout.

- Verb: `watch`
- Users: `system:kube-scheduler`
- Objects: `pods`, `nodes`, `namespaces`
- Namespaces: `none`

1.6.1 Extension-apiserver-authentication

This object is also watched by the Kube Scheduler.

- Verb: `watch`
- Users: `system:kube-scheduler`
- Object: `configmaps/extension-apiserver-authentication`
- Namespaces: `kube-system`

1.7 Nodes watching objects

Each node in the cluster watches pods (non-namespaced), nodes (themselves), and some configmaps.

1.7.1 Pods

- Verb: `watch`
- Users: `system:node:.*`, groups `["system:nodes", "system:authenticated"]`
- Objects: `pods`
- Namespaces: `none`

1.7.2 Nodes

- Verb: `watch`
- Users: `system:node:.*`, groups `["system:nodes", "system:authenticated"]`
- Objects: `nodes/{node-name}`
- Namespaces: `none`

1.7.3 ConfigMaps

Every node hosting some Pod of any namespace will be in charge of watching the serviceaccounts of the Pods that are running.

The CMs being watched are `kube-flannel-cfg`, `kube-proxy`, `kube-root-ca.crt`, and `coredns`. They of course depend on the components deployed in the cluster. Let's keep this generic.

- Verb: `watch`
- Users: `system:node:.*`, groups `["system:nodes","system:authenticated"]`
- Objects: `configmaps/kube-system/configmaps/{serviceaccount}`
- Namespace: `kube-system`

1.8 Nodes creating tokens

1.8.1 Nodes creating tokens for SAs in the kube-system namespace

Since nodes watch the configmaps of the Pods they are running, as the tokens of their service accounts expire, they will recreate them.

- Verb: `create`
- Users: `system:node:.*`, groups `["system:nodes","system:authenticated"]`
- Objects: `/api/v1/namespaces/kube-system/serviceaccounts/{serviceaccount}/token`

1.9 Nodes getting their own status

Nodes poll their own status every ten seconds.

- Verb: `get`
- Users: `system:node:.*`, groups `["system:nodes","system:authenticated"]`
- Objects: `/api/v1/nodes/{the same node}`

1.10 Nodes patching their status to update conditions

Nodes update their status (e.g., `MemoryPressure`, `DiskPressure`, `PIDPressure`) by patching their own status every five minutes.

- Verb: `patch`
- Users: `system:node:.*`, groups `["system:nodes","system:authenticated"]`
- Objects: `/api/v1/nodes/{the same node}`

1.11 CoreDNS watching namespaces

- Verb: `watch`
- Users: `system:serviceaccount:kube-system:coredns`
- Objects: `namespaces`

1.12 Kube-proxy watching nodes

- Verb: watch
- Users: `system:serviceaccount:kube-system:kube-proxy`
- Objects: nodes

2 Log setups level 1

2.1 Namespace log N1 - Simple namespace creation and deletion

- Create a namespace:

```
kubectl create namespace rising
```

- List the namespaces:

```
kubectl get namespaces
```

- Get the namespace using `-o yaml`:

```
kubectl get namespace rising -o yaml
```

- Describe the namespace:

```
kubectl describe namespace rising
```

- Delete the namespace:

```
kubectl delete namespace rising
```

2.2 Namespace log N2 - Namespace with resource quotas

- **Prerequisite:** recreate the rising namespace:

```
kubectl create namespace rising
```

- Create a ResourceQuota for the namespace:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: ResourceQuota
  metadata:
    name: rising-quota
  spec:
    hard:
      pods: "1"
      requests.cpu: "1"
      requests.memory: 1Gi
      limits.cpu: "2"
      limits.memory: 2Gi
EOF
```

- Inspect the ResourceQuota:

```
kubectl -n rising describe resourcequota rising-quota
```

- Create a Pod that will exceed the quota:

```
cat <<EOF | kubectl apply -f -
apiVersion: v1
kind: Pod
metadata:
  name: over-quota-pod
  namespace: rising
spec:
  containers:
  - name: over-quota-container
    image: nginx
    resources:
      requests:
        cpu: "1"
        memory: 1.5Gi
      limits:
        cpu: "2"
        memory: 2Gi
    command: ["sh", "-c", "echo Hello, Kubernetes! && sleep 3600"]
EOF
```

- Now create a Pod that will respect the quota:

```
cat <<EOF | kubectl apply -f -
apiVersion: v1
kind: Pod
metadata:
  name: within-quota-pod
  namespace: rising
spec:
  containers:
  - name: within-quota-container
    image: nginx
    resources:
      requests:
        cpu: "1"
        memory: 1Gi
      limits:
        cpu: "2"
        memory: 2Gi
    command: ["sh", "-c", "echo Hello, Kubernetes! && sleep 3600"]
EOF
```

- Run a further Pod, it will fail:

```
kubectl -n rising run --image=nginx nginx
```

- Clean everything up, one pod will fail because it was never created:

```
kubectl -n rising delete pod over-quota-pod
kubectl -n rising delete pod within-quota-pod
kubectl delete resourcequota rising-quota
```

2.3 Pod log P1 - Simple Pod creation and inspection

- Create a pod:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: rising-pod
    namespace: rising
  spec:
    containers:
      - name: first-container
        image: nginx
      - name: second-container
        image: alpine
        command: ["sh", "-c", "echo Hello, Kubernetes! && sleep 3600"]
EOF
```

- Inspect the Pod:

```
kubectl -n rising describe pod rising-pod
```

- Patch the Pod and try to add a third container, this will fail:

```
kubectl -n rising patch pod rising-pod --type='json' -p='[{"op": "add", "path":
↳ "/spec/containers/1", "value": {"name": "third-container", "image": "nginx"}}]'
```

- Get the pod log:

```
kubectl -n rising logs rising-pod -c second-container
```

- Execute commands in the second containers, the first will complain about the shell, the second will work:

```
kubectl -n rising exec -t rising-pod -c second-container -- /bin/bash -c "echo Hello again,
↳ Kubernetes!"
kubectl -n rising exec -t rising-pod -c second-container -- /bin/sh -c "echo Hello again,
↳ Kubernetes!"
```

- Delete the pod:

```
kubectl -n rising delete pod rising-pod
```

2.4 Pod log P2 - Image download

- **Prerequisite:** run the following helper script to wipe the image cache on all nodes. You **must** create the file `scripts/cluster-setup.sshconfig` with the SSH configuration for the nodes. Refer to the example in the same directory to get started.

```
for node in $(kubectl get nodes -o jsonpath='{.items[*].metadata.name}'); do
  echo "Pruning images on $node"
  ssh -F scripts/cluster-setup.sshconfig $node "sudo crictl rmi --prune"
done
```

- Create a Pod with a container that uses a non-existent image:


```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: image-pod
    namespace: rising
  spec:
    containers:
      - name: image-container
        image: www.fbk.eu/non-existent-image
        command: ["sh", "-c", "echo Hello, Kubernetes! && sleep 3600"]
EOF
```

- Get the pod log:

```
kubectl -n rising logs image-pod -c image-container
```

- Get the events that led to the failure:

```
kubectl get events -n rising --sort-by='.metadata.creationTimestamp' --selector
↳ 'involvedObject.name=image-pod'
```

- Delete the pod:

```
kubectl -n rising delete pod image-pod
```

- Create a pod with an image that will surely not be in cache:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: image-pod
    namespace: rising
  spec:
    containers:
      - name: image-container
        image: busybox
        command: ["sh", "-c", "echo Hello, Kubernetes! && sleep 3600"]
EOF
```

- Wait for the Pod to be downloaded and monitor the events:

```
kubectl wait --for=condition=Ready pod/image-pod -n rising
kubectl get events -n rising --sort-by='.metadata.creationTimestamp'
```

- Delete the pod:

```
kubectl -n rising delete pod image-pod
```

2.5 ConfigMap C1 - Create a ConfigMap and use it in a Pod, then delete it

- Create a ConfigMap:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: ConfigMap
  metadata:
    name: my-config
    namespace: rising
  data:
    status: "decent"
EOF
```

- Get the configmap

```
kubectl -n rising get configmap my-config -o yaml
```

- Deploy a Pod that uses the ConfigMap:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: config-pod
    namespace: rising
  spec:
    containers:
      - name: config-container
        image: alpine
        command: ["sh", "-c", 'echo Status is \${STATUS} && sleep 3600']
        env:
          - name: STATUS
            valueFrom:
              configMapKeyRef:
                name: my-config
                key: status
EOF
```

- Access the logs of the Pod:

```
kubectl wait --for=condition=Ready pod/config-pod -n rising
kubectl -n rising logs config-pod -c config-container
```

- Clean everything up:

```
kubectl -n rising delete pod config-pod
kubectl -n rising delete configmap my-config
```

2.6 Secret S1 - Create a Secret and use it as environment in a Pod, then delete it

- Create a secret

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Secret
  metadata:
    name: my-secret
    namespace: rising
  type: Opaque
  data:
    username: $(echo -n "admin" | base64)
    password: $(echo -n "password" | base64)
EOF
```

- Inspect the secret:

```
kubectl -n rising get secret my-secret -o yaml
```

- Attach the secret to a Pod:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: secret-pod
    namespace: rising
  spec:
    containers:
      - name: secret-container
        image: alpine
        command: ["sh", "-c", 'echo Username is \${USERNAME} && echo Password is \${PASSWORD} && sleep 3600']
        env:
          - name: USERNAME
            valueFrom:
              secretKeyRef:
                name: my-secret
                key: username
          - name: PASSWORD
            valueFrom:
              secretKeyRef:
                name: my-secret
                key: password
EOF
```

- Inspect the Pod:

```
kubectl -n rising describe pod secret-pod
```

- Access the logs of the Pod:

```
kubectl wait --for=condition=Ready pod/secret-pod -n rising
kubectl -n rising logs secret-pod -c secret-container
```

- Clean everything up:

```
kubectl -n rising delete pod secret-pod
kubectl -n rising delete secret my-secret
```

2.7 Secret S2 - Create a Secret and mount it in a Pod

- Create a secret

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Secret
  metadata:
    name: my-secret-2
    namespace: rising
  type: Opaque
  data:
    username: $(echo -n "admin" | base64)
    password: $(echo -n "password" | base64)
EOF
```

- Create a Pod, but now, mount the secret as a volume:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: secret-pod-2
    namespace: rising
  spec:
    containers:
      - name: secret-container-2
        image: alpine
        command: ["sh", "-c", 'echo Username is \${USERNAME} && echo Password is \${PASSWORD} && sleep 3600']
        volumeMounts:
          - name: secret-volume
            mountPath: /etc/secret
    volumes:
      - name: secret-volume
        secret:
          secretName: my-secret-2
EOF
```

- Inspect the Pod:

```
kubectl -n rising describe pod secret-pod-2
```

- Access the Secret from the Pod:

```
kubectl wait --for=condition=Ready pod/secret-pod-2 -n rising
kubectl -n rising exec secret-pod-2 -- cat /etc/secret/username
echo
kubectl -n rising exec secret-pod-2 -- cat /etc/secret/password
```

- Clean everything up:

```
kubectl -n rising delete pod secret-pod-2
kubectl -n rising delete secret my-secret-2
```

2.8 Node log N01 - Node inspection

- Assess the available nodes in the cluster:

```
kubectl get nodes
```

- Inspect one of the nodes:

```
kubectl describe node kubeadm-worker1
```

2.9 Certificate log CSR1 - Create a CertificateSigningRequest and approve it

- Note: an helper script is available in `scripts/generate_k8s_user.sh` that automates what is done in this section.
- Use openssl to generate a key and a certificate signing request:

```
export TMPDIR=$(mktemp -d)
export ORIGINDIR=$(pwd)
cd $TMPDIR
export KUBEUSER=malicious-user
openssl genpkey -out ${KUBEUSER}.key -algorithm Ed25519
```

```
openssl req -new -key ${KUBEUSER}.key -out ${KUBEUSER}.csr -subj
↳ "/CN=${KUBEUSER}/O=rising/O=fbk/"
echo "Key created, available in ${TMPDIR} for user ${KUBEUSER}"
```

- Create a CertificateSigningRequest:

```
cat <<EOF | kubectl apply -f -
  apiVersion: certificates.k8s.io/v1
  kind: CertificateSigningRequest
  metadata:
    name: ${KUBEUSER}
  spec:
    request: $(cat ${KUBEUSER}.csr | base64 | tr -d '\n')
    signerName: kubernetes.io/kube-apiserver-client
    usages:
      - client auth
EOF
```

- Deny the CertificateSigningRequest:

```
kubectl certificate deny ${KUBEUSER}
```

- Delete the failed CSR and generate a new keypair for a legitimate user:

```
rm ${KUBEUSER}.csr ${KUBEUSER}.key
kubectl delete csr ${KUBEUSER}
export KUBEUSER=rising-user
openssl genpkey -out ${KUBEUSER}.key -algorithm Ed25519
openssl req -new -key ${KUBEUSER}.key -out ${KUBEUSER}.csr -subj
↳ "/CN=${KUBEUSER}/O=rising/O=fbk/"
echo "Key created, available in ${TMPDIR} for user ${KUBEUSER}"
```

- Generate a new CertificateSigningRequest:

```
cat <<EOF | kubectl apply -f -
  apiVersion: certificates.k8s.io/v1
  kind: CertificateSigningRequest
  metadata:
    name: ${KUBEUSER}
  spec:
    request: $(cat ${KUBEUSER}.csr | base64 | tr -d '\n')
    signerName: kubernetes.io/kube-apiserver-client
    usages:
      - client auth
EOF
```

- Inspect the status of the CSR:

```
kubectl get csr ${KUBEUSER} -o yaml
```

- Approve the CertificateSigningRequest:

```
kubectl certificate approve ${KUBEUSER}
```

- Obtain the certificate from the CSR and store it in the folder

```
kubectl get csr ${KUBEUSER} -o jsonpath='{.status.certificate}' | base64 -d > ${KUBEUSER}.cert
head -n 2 ${KUBEUSER}.cert
```

- Use the certificate to generate a .kubeconfig for the user:

```
cat <<EOF > ${KUBEUSER}.kubeconfig
  apiVersion: v1
  kind: Config
  preferences: {}
  clusters:
    $(kubectl config view --raw --minify --flatten | yq .clusters)
  contexts: []
  users: []
EOF
kubectl --kubeconfig ${KUBEUSER}.kubeconfig config set-credentials ${KUBEUSER}
↪ --client-certificate=${KUBEUSER}.crt --client-key=${KUBEUSER}.key --embed-certs=true
kubectl --kubeconfig ${KUBEUSER}.kubeconfig config set-context ${KUBEUSER}
↪ --cluster=kubernetes --user=${KUBEUSER} --cluster=risinglab
kubectl --kubeconfig ${KUBEUSER}.kubeconfig config use-context ${KUBEUSER}
cp ${KUBEUSER}.kubeconfig ${ORIGINDIR}
```

2.10 Role log R1 - Create a Role and bind it to a User

- **Prerequisite:** complete CSR1 and obtain the credentials for the user. Make sure they are available as rising-user.kubeconfig in the current directory.
- Create a Role:

```
cat <<EOF | kubectl apply -f -
  apiVersion: rbac.authorization.k8s.io/v1
  kind: Role
  metadata:
    name: rising-reader
    namespace: rising
  rules:
  - apiGroups: [""]
    resources: ["pods"]
    verbs: ["get", "list"]
EOF
```

- List the roles in the namespace:

```
kubectl -n rising get roles
```

- We changed idea: let's also give get, list, patch and watch for Secrets:

```
kubectl patch role rising-reader -n rising --type='json' -p='[{"op": "add", "path":
↪ "/rules/0", "value": {"apiGroups": [""], "resources": ["secrets"], "verbs": ["get",
↪ "list", "patch", "watch"]}]}'
```

- Let's create a RoleBinding and assign it to the user:

```
cat <<EOF | kubectl apply -f -
  apiVersion: rbac.authorization.k8s.io/v1
  kind: RoleBinding
  metadata:
    name: pod-reader-binding
    namespace: rising
  subjects:
  - kind: User
    name: rising-user
  roleRef:
    kind: Role
    name: rising-reader
    apiGroup: rbac.authorization.k8s.io
EOF
```

- Test the permissions of the user:

```
kubectl --kubeconfig rising-user.kubeconfig get pods -n rising
kubectl --kubeconfig rising-user.kubeconfig get secrets -n rising
kubectl --kubeconfig rising-user.kubeconfig delete pod non-existent-pod -n rising
```

2.11 Role log R2 - Create a ClusterRole and bind it to a Group

- **Prerequisite:** complete CSR1 and obtain the credentials for the user. Make sure they are available as `rising-user.kubeconfig` in the current directory.
- **Prerequisite:** complete R1 and keep the `rising-reader` Role.
- Create a ClusterRole:

```
cat <<EOF | kubectl apply -f -
  apiVersion: rbac.authorization.k8s.io/v1
  kind: ClusterRole
  metadata:
    name: cluster-reader
  rules:
  - apiGroups: [""]
    resources: ["pods"]
    verbs: ["get", "list"]
  - apiGroups: [""]
    resources: ["secrets"]
    verbs: ["get", "list", "patch", "watch"]
EOF
```

- Actually, let's straight out replace the ClusterRole with a new one:

```
cat <<EOF | kubectl apply -f -
  apiVersion: rbac.authorization.k8s.io/v1
  kind: ClusterRole
  metadata:
    name: cluster-reader
  rules:
  - apiGroups: [""]
    resources: ["nodes"]
    verbs: ["get", "list"]
EOF
```

- Create a ClusterRoleBinding and assign it to the `fbk` group:

```
cat <<EOF | kubectl apply -f -
  apiVersion: rbac.authorization.k8s.io/v1
  kind: ClusterRoleBinding
  metadata:
    name: cluster-reader-binding
  subjects:
  - kind: Group
    name: fbk
  roleRef:
    kind: ClusterRole
    name: cluster-reader
    apiGroup: rbac.authorization.k8s.io
EOF
```

- Test the permissions of the user:

```
kubectl --kubeconfig rising-user.kubeconfig get nodes
kubectl --kubeconfig rising-user.kubeconfig get pods -n rising
kubectl --kubeconfig rising-user.kubeconfig get configmaps -n rising
kubectl --kubeconfig rising-user.kubeconfig get pods -n kube-system
```

- Clean everything up:

```
kubectl delete rolebinding pod-reader-binding -n rising
kubectl delete role rising-reader -n rising
kubectl delete clusterrolebinding cluster-reader-binding
kubectl delete clusterrole cluster-reader
kubectl delete csr rising-user
```

2.12 Role log R3 - Create a ServiceAccount for a long-running Pod and bind it to a Role

- Set up a ServiceAccount:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: ServiceAccount
  metadata:
    name: long-running-sa
    namespace: rising
EOF
```

- Create a Role:

```
cat <<EOF | kubectl apply -f -
  apiVersion: rbac.authorization.k8s.io/v1
  kind: Role
  metadata:
    name: long-running-role
    namespace: rising
  rules:
  - apiGroups: [""]
    resources: ["pods"]
    verbs: ["get", "list"]
EOF
```

- Bind the ServiceAccount to the Role:

```
cat <<EOF | kubectl apply -f -
  apiVersion: rbac.authorization.k8s.io/v1
  kind: RoleBinding
  metadata:
    name: long-running-role-binding
    namespace: rising
  subjects:
  - kind: ServiceAccount
    name: long-running-sa
    namespace: rising
  roleRef:
    kind: Role
    name: long-running-role
    apiGroup: rbac.authorization.k8s.io
EOF
```

- Create a Pod that uses the ServiceAccount:


```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: long-running-pod
    namespace: rising
  spec:
    serviceAccountName: long-running-sa
    containers:
    - name: long-running-container
      image: ubuntu
      command: ["sh", "-c", "echo Hello, Kubernetes! && sleep 3600"]
EOF
```

- Inspect the Pod:

```
kubectl -n rising describe pod long-running-pod | head -n 10
```

- Try to use the ServiceAccount from inside the Pod:

```
kubectl wait --for=condition=Ready pod/long-running-pod -n rising
kubectl -n rising exec long-running-pod -- sh -c 'apt-get -qq update; apt-get install -yqq
  curl yq >/dev/null;
  export KUBE_TOKEN=$(cat /var/run/secrets/kubernetes.io/serviceaccount/token);
  curl -sSk -H "Authorization: Bearer $KUBE_TOKEN"
  "https://$KUBERNETES_SERVICE_HOST:$KUBERNETES_PORT_443_TCP_PORT/apis/authentication.k8s.io/v1/selfsubject
> temp.json; jq .message temp.json
  curl -sSk -H "Authorization: Bearer $KUBE_TOKEN"
  ↪ "https://$KUBERNETES_SERVICE_HOST:$KUBERNETES_PORT_443_TCP_PORT/api/v1/namespaces/rising/pods"
  ↪ > temp.json; head -n 5 temp.json'
```

- Clean everything up:

```
kubectl -n rising delete pod long-running-pod
kubectl -n rising delete role long-running-role
kubectl -n rising delete rolebinding long-running-role-binding
kubectl -n rising delete serviceaccount long-running-sa
```

2.13 Access Review AR1 - Use can-i to view permissions of self, other users, over different namespaces

- Create a set of ServiceAccounts for the logs (while this should not be the case in a real-world scenario, it is done since SAs and Users are equivalent once authorized):

```
kubectl apply -f - <<EOF
  apiVersion: v1
  kind: ServiceAccount
  metadata:
    name: alice
    namespace: rising
  ---
  apiVersion: v1
  kind: ServiceAccount
  metadata:
    name: bob
    namespace: rising
  ---
  apiVersion: v1
  kind: ServiceAccount
  metadata:
    name: charlie
    namespace: rising
EOF
```

- Create tokens for the ServiceAccounts:

```
kubectl apply -f - <<EOF
  apiVersion: v1
  kind: Secret
  metadata:
    name: alice-token
    namespace: rising
    annotations:
      kubernetes.io/service-account.name: alice
  type: kubernetes.io/service-account-token
  ---
  apiVersion: v1
  kind: Secret
  metadata:
    name: bob-token
    namespace: rising
    annotations:
      kubernetes.io/service-account.name: bob
  type: kubernetes.io/service-account-token
  ---
  apiVersion: v1
  kind: Secret
  metadata:
    name: charlie-token
    namespace: rising
    annotations:
      kubernetes.io/service-account.name: charlie
  type: kubernetes.io/service-account-token
  ---
EOF
```

- Create two Roles, one for reading Pods and one for reading Secrets:

```
kubectl apply -f - <<EOF
  apiVersion: rbac.authorization.k8s.io/v1
  kind: Role
  metadata:
    name: pod-reader
    namespace: rising
  rules:
  - apiGroups: [""]
    resources: ["pods"]
    verbs: ["get", "list"]
  ---
  apiVersion: rbac.authorization.k8s.io/v1
  kind: Role
  metadata:
    name: secret-reader
    namespace: rising
  rules:
  - apiGroups: [""]
    resources: ["secrets"]
    verbs: ["get", "list"]
EOF
```

- Bind the ServiceAccounts to the Roles:

```
kubectl apply -f - <<EOF
  apiVersion: rbac.authorization.k8s.io/v1
  kind: RoleBinding
  metadata:
    name: pod-reader-binding
    namespace: rising
  subjects:
  - kind: ServiceAccount
    name: alice
    namespace: rising
  - kind: ServiceAccount
    name: bob
    namespace: rising
  roleRef:
    kind: Role
    name: pod-reader
    apiGroup: rbac.authorization.k8s.io
  ---
  apiVersion: rbac.authorization.k8s.io/v1
  kind: RoleBinding
  metadata:
    name: secret-reader-binding
    namespace: rising
  subjects:
  - kind: ServiceAccount
    name: charlie
    namespace: rising
  roleRef:
    kind: Role
    name: secret-reader
    apiGroup: rbac.authorization.k8s.io
EOF
```

- Obtain credentials for the ServiceAccounts and generate a kubeconfig for them:

```
kubectl get secret alice-token -n rising -o jsonpath='{.data.token}' | base64 -d > alice.token
kubectl get secret bob-token -n rising -o jsonpath='{.data.token}' | base64 -d > bob.token
kubectl get secret charlie-token -n rising -o jsonpath='{.data.token}' | base64 -d >
↵ charlie.token
```

```

cat <<EOF > alice.kubeconfig
  apiVersion: v1
  kind: Config
  preferences: {}
  clusters:
  $(kubectl config view --raw --minify --flatten | yq .clusters)
  contexts: []
  users: []
EOF
cp alice.kubeconfig bob.kubeconfig
cp alice.kubeconfig charlie.kubeconfig
kubectl --kubeconfig alice.kubeconfig config set-credentials alice --token=$(cat alice.token)
kubectl --kubeconfig bob.kubeconfig config set-credentials bob --token=$(cat bob.token)
kubectl --kubeconfig charlie.kubeconfig config set-credentials charlie --token=$(cat
↳ charlie.token)
kubectl --kubeconfig alice.kubeconfig config set-context alice --cluster=kubernetes
↳ --user=alice --cluster=risinglab
kubectl --kubeconfig bob.kubeconfig config set-context bob --cluster=kubernetes --user=bob
↳ --cluster=risinglab
kubectl --kubeconfig charlie.kubeconfig config set-context charlie --cluster=kubernetes
↳ --user=charlie --cluster=risinglab
kubectl --kubeconfig alice.kubeconfig config use-context alice
kubectl --kubeconfig bob.kubeconfig config use-context bob
kubectl --kubeconfig charlie.kubeconfig config use-context charlie
rm alice.token bob.token charlie.token

```

- Check as Alice if she can list Pods and Secrets in the namespace:

```

kubectl --kubeconfig alice.kubeconfig auth can-i list pods -n rising
kubectl --kubeconfig alice.kubeconfig auth can-i list secrets -n rising

```

- Check as Bob if he can list Pods in the namespaces rising and kube-system:

```

kubectl --kubeconfig bob.kubeconfig auth can-i list pods -n rising
kubectl --kubeconfig bob.kubeconfig auth can-i list pods -n kube-system

```

- Check as Charlie if he can list Secrets in the namespace:

```

kubectl --kubeconfig charlie.kubeconfig auth can-i list secrets -n rising

```

- Check as Charlie if he can list nodes:

```

kubectl --kubeconfig charlie.kubeconfig auth can-i list nodes

```

- Check what Charlie can do in the rising namespace:

```

kubectl --kubeconfig charlie.kubeconfig auth can-i --list -n rising

```

2.14 Access Review AR2 - Use auth whoami to view the current user

- **Prerequisite:** complete AR1 and obtain the credentials for the users. Make sure they are available as `alice.kubeconfig`, `bob.kubeconfig` and `charlie.kubeconfig` in the current directory.
- Check as each of the users who they are:

```

kubectl --kubeconfig alice.kubeconfig auth whoami
kubectl --kubeconfig bob.kubeconfig auth whoami
kubectl --kubeconfig charlie.kubeconfig auth whoami

```

2.15 Access Review AR3 - Impersonating a user

- **Prerequisite:** complete AR1 and obtain the credentials for the users. Make sure they are available as `alice.kubeconfig`, `bob.kubeconfig` and `charlie.kubeconfig` in the current directory.
- Augment Charlie's capabilities to let him impersonate ServiceAccounts:

```
kubectl apply -f - <<EOF
  apiVersion: rbac.authorization.k8s.io/v1
  kind: ClusterRole
  metadata:
    name: impersonator
  rules:
    - apiGroups: [""]
      resources: ["serviceaccounts"]
      verbs: ["impersonate"]
      resourceNames: ["alice", "bob"]
  ---
  apiVersion: rbac.authorization.k8s.io/v1
  kind: ClusterRoleBinding
  metadata:
    name: impersonator-binding
  subjects:
    - kind: ServiceAccount
      name: charlie
      namespace: rising
  roleRef:
    kind: ClusterRole
    name: impersonator
    apiGroup: rbac.authorization.k8s.io
EOF
```

- Check if Charlie can impersonate any user or SA:

```
kubectl --kubeconfig charlie.kubeconfig auth can-i impersonate users
kubectl --kubeconfig charlie.kubeconfig auth can-i impersonate serviceaccounts
```

- Let Charlie see if Alice can list Pods in `rising`, and what she can do in `kube-system`:

```
kubectl --kubeconfig charlie.kubeconfig auth can-i list pods -n rising --as
↳ system:serviceaccount:rising:alice
kubectl --kubeconfig charlie.kubeconfig auth can-i --list -n kube-system --as
↳ system:serviceaccount:rising:alice
```

- As Charlie, let's impersonate Bob and create a Pod in the `rising` namespace, then list Pods:

```
kubectl --kubeconfig charlie.kubeconfig --as system:serviceaccount:rising:bob run
↳ --image=ubuntu --restart=Never --namespace=rising --command -- sleep 3600
kubectl --kubeconfig charlie.kubeconfig --as system:serviceaccount:rising:bob get pods -n
↳ rising
```

- As Charlie, let's impersonate Alice and try to list nodes:

```
kubectl --kubeconfig charlie.kubeconfig --as system:serviceaccount:rising:alice get nodes
```

- Clean everything up:

```
kubectl delete clusterrolebinding impersonator-binding
kubectl delete clusterrole impersonator
kubectl delete rolebinding pod-reader-binding -n rising
kubectl delete role pod-reader -n rising
kubectl delete rolebinding secret-reader-binding -n rising
```

```
kubectl delete role secret-reader -n rising
kubectl delete secret alice-token bob-token charlie-token -n rising
kubectl delete serviceaccount alice bob charlie -n rising
rm alice.kubeconfig bob.kubeconfig charlie.kubeconfig
```

3 Log setups level 2

3.1 Deployment log D1 - Create a Deployment for a ReplicaSet and expose it as a Service

- Create a Deployment:

```
cat <<EOF | kubectl apply -f -
  apiVersion: apps/v1
  kind: Deployment
  metadata:
    name: test-deployment
    namespace: rising
    labels:
      app: test
  spec:
    replicas: 2
    selector:
      matchLabels:
        app: test
    template:
      metadata:
        labels:
          app: test
      spec:
        containers:
          - name: nginx
            image: nginx:latest
            ports:
              - containerPort: 80
EOF
```

- Inspect the Deployment:

```
kubectl get deployments test-deployment -n rising
```

- Inspect the rollout status:

```
kubectl rollout status deployment test-deployment -n rising
```

- Inspect the ReplicaSet created by the Deployment:

```
kubectl get replicaset -l app=test -n rising
```

- Describe the ReplicaSet directly:

```
kubectl describe replicaset -l app=test -n rising
```

- Expose the Deployment as a Service:

```
kubectl expose deployment test-deployment --type=NodePort --port=80 --target-port=80
↵ --name=test-deployment -n rising
```

- Inspect the Service:

```
kubectl get services test-deployment -n rising
```

- Tear everything down:

```
kubectl delete service test-deployment -n rising
kubectl delete deployment test-deployment -n rising
```

3.2 Deployment log D2 - Scale a Deployment and update its image

- Create a Deployment again:

```
cat <<EOF | kubectl apply -f -
apiVersion: apps/v1
kind: Deployment
metadata:
  name: test-deployment
  namespace: rising
  labels:
    app: test
spec:
  replicas: 2
  selector:
    matchLabels:
      app: test
  template:
    metadata:
      labels:
        app: test
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
EOF
```

- Scale the Deployment to 3 replicas:

```
kubectl scale deployment test-deployment --replicas=3 -n rising
```

- Inspect the Deployment:

```
kubectl get deployments test-deployment -n rising
```

- Update the image of the Deployment:

```
kubectl set image deployment test-deployment nginx=nginx:1.16.1 -n rising
```

- Inspect the rollout status:

```
kubectl rollout status deployment test-deployment -n rising
```

- Scale down the Deployment to 2 replicas:

```
kubectl scale deployment test-deployment --replicas=2 -n rising
```

- Inspect ReplicaSets:

```
kubectl get replicaset -l app=test -n rising
```

- Inspect the Pods:

```
kubectl get pods -l app=test -n rising
```

- Tear everything down:

```
kubectl delete deployment test-deployment -n rising
```

3.3 Deployment log D3 - Roll back, pause and resume a Deployment

- Create a Deployment again:

```
cat <<EOF | kubectl apply -f -
  apiVersion: apps/v1
  kind: Deployment
  metadata:
    name: test-deployment
    namespace: rising
    labels:
      app: test
  spec:
    replicas: 2
    selector:
      matchLabels:
        app: test
    template:
      metadata:
        labels:
          app: test
      spec:
        containers:
          - name: nginx
            image: nginx:latest
            ports:
              - containerPort: 80
EOF
```

- Wait for the Deployment to be ready:

```
kubectl wait --for=condition=Available deployment/test-deployment -n rising
```

- Pause the Deployment:

```
kubectl rollout pause deployment test-deployment -n rising
```

- Update the image of the Deployment:

```
kubectl set image deployment test-deployment nginx=nginx:1.16.1 -n rising
```

- Inspect the rollout history:

```
kubectl rollout history deployment test-deployment -n rising
```

- Inspect the ReplicaSets:

```
kubectl get replicaset -l app=test -n rising
```

- Resume the Deployment:

```
kubectl rollout resume deployment test-deployment -n rising
```

- Inspect the rollout history again:


```
kubectl rollout history deployment test-deployment -n rising
```

- Roll back the Deployment to the previous revision:

```
kubectl rollout undo deployment test-deployment -n rising
```

- Inspect the rollout history one more time:

```
kubectl rollout history deployment test-deployment -n rising
```

- List Pods:

```
kubectl get pods -l app=test
```

- Tear everything down:

```
kubectl delete deployment test-deployment -n rising
```

3.4 Deployment log D4 - Watch a Deployment fail and recover

- Install a restrictive ResourceQuota:

```
cat <<EOF | kubectl apply -f -
apiVersion: v1
kind: ResourceQuota
metadata:
  name: test-quota
  namespace: rising
spec:
  hard:
    pods: "1"
EOF
```

- Create a Deployment again:

```
cat <<EOF | kubectl apply -f -
apiVersion: apps/v1
kind: Deployment
metadata:
  name: test-deployment
  namespace: rising
  labels:
    app: test
spec:
  replicas: 2
  selector:
    matchLabels:
      app: test
  template:
    metadata:
      labels:
        app: test
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
EOF
```

- Check the status of the Deployment:

```
kubectl describe deployment test-deployment -n rising
```

- Relax the ResourceQuota:

```
kubectl patch resourcequota test-quota -p '{"spec":{"hard":{"pods":"5"}}}' -n rising
```

- Check the status of the Deployment again:

```
kubectl rollout status deployment test-deployment -n rising
```

- Change the image of the Deployment and put a non-existent image:

```
kubectl set image deployment test-deployment nginx=engincs:ahahahah -n rising
```

- Check the status of the Deployment:

```
kubectl describe deployment test-deployment -n rising
```

- List the Pods:

```
kubectl get pods -l app=test -n rising
```

- Fix the image of the Deployment:

```
kubectl set image deployment test-deployment nginx=nginx:1.16.1 -n rising
```

- Check the status again:

```
kubectl get deployment test-deployment -n rising
```

- Now, set an impossible affinity for the Deployment:

```
kubectl patch deployment test-deployment -p  
→ '{"spec":{"template":{"spec":{"affinity":{"nodeAffinity":{"requiredDuringSchedulingIgnoredDuringExecution":{"nodeSelectorTerms":[{"matchExpressions":[{"key":"node.kubernetes.io/instance-type","operator":"NotIn","values":["n1-standard-4"]}]}}}}}}}'  
→ -n rising
```

- Check the status of the Deployment:

```
kubectl describe deployment test-deployment -n rising
```

- Inspect the Pods:

```
kubectl get pods -l app=test -n rising
```

- Tear everything down:

```
kubectl delete deployment test-deployment -n rising  
kubectl delete resourcequota test-quota -n rising
```

3.5 Deployment log D5 - Manage two Deployments contemporaneously

- Create two Deployments together:

```
kubectl apply -f - <<EOF
apiVersion: apps/v1
kind: Deployment
metadata:
  name: interesting-deployment-1
  namespace: rising
  labels:
    app: interesting-app
spec:
  replicas: 4
  selector:
    matchLabels:
      app: interesting-app
  template:
    metadata:
      labels:
        app: interesting-app
    spec:
      containers:
      - name: interesting-container-1
        image: nginx:1.19.6
        ports:
        - containerPort: 80
      - name: sidecar-container-1
        image: busybox:1.31.1
        command: ['sh', '-c', 'echo sidecar-container-1 is running && sleep 3600']
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: interesting-deployment-2
  namespace: rising
  labels:
    app: interesting-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: interesting-app
  template:
    metadata:
      labels:
        app: interesting-app
    spec:
      containers:
      - name: interesting-container-2
        image: nginx:latest
      initContainers:
      - name: init-container-1
        image: busybox:1.31.1
        command: ['sh', '-c', 'echo init-container-1 is running && sleep 5']
EOF
```

- Wait for the Deployments to be ready:

```
kubectl wait --for=condition=Available deployment/interesting-deployment-1 -n rising
kubectl wait --for=condition=Available deployment/interesting-deployment-2 -n rising
```

- Check the rollout status of the Deployments:

```
kubectl rollout status deployment interesting-deployment-1 -n rising
kubectl rollout status deployment interesting-deployment-2 -n rising
```

- Inspect the Deployments:

```
kubectl get deployments -l app=interesting-app -n rising
```

3.6 ReplicaSet log RS1 - Use bare ReplicaSets to manage Pods

- Create a ReplicaSet:

```
cat <<EOF | kubectl apply -f -
  apiVersion: apps/v1
  kind: ReplicaSet
  metadata:
    name: frontend
    namespace: rising
    labels:
      app: guestbook
      tier: frontend
  spec:
    replicas: 3
    selector:
      matchLabels:
        tier: frontend
    template:
      metadata:
        labels:
          tier: frontend
      spec:
        containers:
          - name: outward
            image: httpd:latest
EOF
```

- List the ReplicaSets:

```
kubectl get replicaset -n rising
```

- Inspect the ReplicaSet:

```
kubectl describe replicaset frontend -n rising
```

- Scale down the ReplicaSet:

```
kubectl scale replicaset frontend --replicas=2 -n rising
```

- Inspect the Pods:

```
kubectl get pods -l tier=frontend -n rising
```

- Tear everything down:

```
kubectl delete replicaset frontend -n rising
```

3.7 DaemonSet log DS1 - Manage a DaemonSet

- Create a DaemonSet:

```
cat <<EOF | kubectl apply -f -
  apiVersion: apps/v1
  kind: DaemonSet
  metadata:
    name: fluentd-elasticsearch
    namespace: kube-system
    labels:
      k8s-app: fluentd-logging
  spec:
    selector:
      matchLabels:
        name: fluentd-elasticsearch
    template:
      metadata:
        labels:
          name: fluentd-elasticsearch
      spec:
        tolerations:
          - key: node-role.kubernetes.io/control-plane
            operator: Exists
            effect: NoSchedule
        containers:
          - name: fluentd-elasticsearch
            image: quay.io/fluentd_elasticsearch/fluentd:v2.5.2
            resources:
              limits:
                memory: 200Mi
              requests:
                cpu: 100m
                memory: 200Mi
            volumeMounts:
              - name: varlog
                mountPath: /var/log
        terminationGracePeriodSeconds: 30
        volumes:
          - name: varlog
            hostPath:
              path: /var/log
EOF
```

- Inspect the DaemonSet:

```
kubectl get daemonsets fluentd-elasticsearch -n kube-system
```

- Inspect the Pods:

```
kubectl get pods -l name=fluentd-elasticsearch -n kube-system
```

- Cordon kubeadm-worker1:

```
kubectl cordon kubeadm-worker1
```

- Check the Pods again:

```
kubectl get pods -l name=fluentd-elasticsearch -n kube-system
```

- Uncordon and add a label to kubeadm-worker1:

```
kubectl uncordon kubeadm-worker1
kubectl taint nodes kubeadm-worker1 test-label=no-schedule:NoExecute
```

- Edit the DaemonSet to tolerate the taint, this will drop out the master node:

```
kubectl patch daemonset fluentd-elasticsearch -p
↪ '{"spec":{"template":{"spec":{"tolerations":[{"key":"test-label","operator":"Equal","effect":"NoExecute"}]}}}'
↪ -n kube-system
```

- Check the Pods again:

```
kubectl get pods -l name=fluentd-elasticsearch -n kube-system
```

- Clean everything up:

```
kubectl delete daemonset fluentd-elasticsearch -n kube-system
kubectl taint nodes kubeadm-worker1 test-label-
```

3.8 Job log J1 - Manage a Job

- Create a Job:

```
cat <<EOF | kubectl apply -f -
apiVersion: batch/v1
kind: Job
metadata:
  name: pi
  namespace: rising
spec:
  template:
    spec:
      containers:
      - name: pi
        image: perl:5.34.0
        command: ["perl", "-Mbignum=bpi", "-wle", "print bpi(2000)"]
        restartPolicy: Never
      backoffLimit: 4
EOF
```

- List available Jobs:

```
kubectl get jobs -n rising
```

- Inspect the Job:

```
kubectl describe jobs pi -n rising
```

- Check the Pods:

```
kubectl get pods -l job-name=pi -n rising
```

- Assess the logs generated by the Job:

```
kubectl wait --for=condition=complete job/pi -n rising
kubectl logs jobs/pi -n rising | head -c 20
```

- Update the spec of the Job to trigger auto deletion:

```
kubectl patch job pi -p '{"spec":{"ttlSecondsAfterFinished":60}}' -n rising
```

- Wait for the Job to be deleted:

```
kubectl wait --for=delete job/pi -n rising --timeout=80s
```

3.9 CronJob log CJ1 - Manage a CronJob

- Create a CronJob:

```
cat <<EOF | kubectl apply -f -
  apiVersion: batch/v1
  kind: CronJob
  metadata:
    name: hello
    labels:
      cj: hello
    namespace: rising
  spec:
    schedule: "* * * * *"
    jobTemplate:
      metadata:
        labels:
          cj: hello
      spec:
        template:
          spec:
            containers:
            - name: hello
              image: busybox:1.28
              imagePullPolicy: IfNotPresent
              command:
              - /bin/sh
              - -c
                date; echo Hello from the Kubernetes cluster
            restartPolicy: OnFailure
EOF
```

- List available CronJobs:

```
kubectl get cronjobs -n rising
```

- Inspect the CronJob:

```
kubectl describe cronjobs hello -n rising
```

- Wait two-three minutes and check the Jobs created by the CronJob:

```
sleep 120
kubectl get jobs -l cj=hello -n rising
```

- Clean everything up:

```
kubectl delete cronjob hello -n rising
```

4 Log setups level 3

4.1 Pod log P3 - Use kubectl debug and Ephemerals Containers

- Run a dumb Pod that does not have a shell:

```
kubectl run -n rising --restart=Never --image=registry.k8s.io/pause:latest dumb-pod -- sleep
↪ 3600
```

- Try to launch a `kubectl exec`, it will fail:

```
kubectl exec -it dumb-pod -- sh
```

- Use `kubectl debug` to attach an Ephemeral Container:

```
kubectl debug dumb-pod --image=busybox:1.31.1 --target=dumb-pod
```

- Describe the newly created Ephemeral Container:

```
kubectl describe pod dumb-pod -n rising
```

- Destroy the ephemeral Pod and the dumb Pod:

```
kubectl delete pod dumb-pod -n rising
```

4.2 Pod log P4 - Use Port Forwarding to access a Pod

- Create a Deployment:

```
cat <<EOF | kubectl apply -f -
apiVersion: apps/v1
kind: Deployment
metadata:
  name: mongo
  labels:
    app.kubernetes.io/name: mongo
    app.kubernetes.io/component: backend
spec:
  selector:
    matchLabels:
      app.kubernetes.io/name: mongo
      app.kubernetes.io/component: backend
  replicas: 1
  template:
    metadata:
      labels:
        app.kubernetes.io/name: mongo
        app.kubernetes.io/component: backend
    spec:
      containers:
      - name: mongo
        image: mongo:4.2
        args:
          - --bind_ip
          - 0.0.0.0
        resources:
          requests:
            cpu: 100m
            memory: 100Mi
        ports:
          - containerPort: 27017
EOF
```

- Inspect Deployments, ReplicaSets, Pods:

```
kubectl get deployments -n rising
kubectl get replicaset -n rising
kubectl get pods -n rising
```

- Create a Service to expose Mongo:


```
cat <<EOF | kubectl apply -f -
apiVersion: v1
kind: Service
metadata:
  name: mongo
  labels:
    app.kubernetes.io/name: mongo
    app.kubernetes.io/component: backend
spec:
  ports:
    - port: 27017
      targetPort: 27017
  selector:
    app.kubernetes.io/name: mongo
    app.kubernetes.io/component: backend
EOF
```

- Port-forward the Pod and send the process to background:

```
kubectl port-forward deployment/mongo 28015:27017 &
pid=$!
```

- Clean everything up:

```
kill $pid
kubectl delete deployment mongo -n rising
kubectl delete service mongo -n rising
```

4.3 LimitRange log LR1 - Create a LimitRange to restrict the resources of a Pod

- Define a LimitRange for the namespace:

```
cat <<EOF | kubectl apply -f -
apiVersion: v1
kind: LimitRange
metadata:
  name: limit-range
  namespace: rising
spec:
  limits:
    - type: Container
      default:
        cpu: 300m
        memory: 200Mi
      defaultRequest:
        cpu: 200m
        memory: 100Mi
      max:
        cpu: 500m
        memory: 500Mi
      min:
        cpu: 100m
        memory: 50Mi
EOF
```

- Inspect the newly created LimitRange:

```
kubectl describe limitrange limit-range -n rising
```

- Create a Pod that does not fit the LimitRange:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: bad-pod
    namespace: rising
  spec:
    containers:
      - name: bad-container
        image: busybox:1.31.1
        command: ["sh", "-c", "sleep 3600"]
        resources:
          requests:
            cpu: 600m
            memory: 300Mi
EOF
```

- Create a lawful Pod:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: good-pod
    namespace: rising
  spec:
    containers:
      - name: good-container
        image: busybox:1.31.1
        command: ["sh", "-c", "sleep 3600"]
        resources:
          requests:
            cpu: 200m
            memory: 100Mi
EOF
```

- Inspect the good Pod:

```
kubectl describe pod good-pod -n rising
```

- Delete it:

```
kubectl delete pod good-pod -n rising
```

- Clean everything up:

```
kubectl delete limitrange limit-range -n rising
```

4.4 Service log SV1 - Expose the three types of Services from a Pod

- Create a Pod:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: service-pod
    namespace: rising
    labels:
      app: service-app
  spec:
    containers:
      - name: service-container
        image: nginx:1.19.6
        ports:
          - containerPort: 80
            name: http
EOF
```

- Expose a ClusterIP service (by default if you don't specify its type):

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Service
  metadata:
    name: nginx-service
    namespace: rising
  spec:
    selector:
      app: service-app
    ports:
      - name: http-external
        protocol: TCP
        port: 80
        targetPort: http
EOF
```

- Inspect the newly create Service:

```
kubectl describe service nginx-service -n rising
```

- List the Services:

```
kubectl get services -n rising
```

- Update the Service, and replace it with a NodePort:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Service
  metadata:
    name: nginx-service
    namespace: rising
  spec:
    type: NodePort
    selector:
      app: service-app
    ports:
      - name: http-external
        protocol: TCP
        port: 80
        targetPort: http
        nodePort: 30080
EOF
```

- Inspect the updated Service:

```
kubectl describe service nginx-service -n rising
```

- List Services again:

```
kubectl get services -n rising
```

LoadBalancer services are not supported in my environment, so I can't test them.

- Add an ExternalName service:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Service
  metadata:
    name: external-service
    namespace: rising
  spec:
    type: ExternalName
    externalName: risinglab.fbk.eu
EOF
```

- Clean all the resources:

```
kubectl delete pod service-pod -n rising
kubectl delete service nginx-service -n rising
kubectl delete service external-service -n rising
```

4.5 Service log SV2 - Expose Deployments through Services

- Create a Deployment with an explicit containerPort:

```
cat <<EOF | kubectl apply -f -
  apiVersion: apps/v1
  kind: Deployment
  metadata:
    name: exposed-deployment
    namespace: rising
  spec:
    selector:
      matchLabels:
        run: nginx-deployment
    replicas: 2
    template:
      metadata:
        labels:
          run: nginx-deployment
      spec:
        containers:
          - name: nginx-runner
            image: nginx
            ports:
              - containerPort: 80
EOF
```

- Check the newly created Deployment:

```
kubectl get deployments -n rising -o wide
```

- Expose the Deployment, it will default to a ClusterIP using the only containerPort available:

```
kubectl expose deployment exposed-deployment
```

- List the Services:

```
kubectl get services -n rising
```

- Describe the Service:

```
kubectl describe service exposed-deployment -n rising
```

- List the EndpointSlices:

```
kubectl get endpointslices -n rising
```

- Scale the Deployment:

```
kubectl scale deployment exposed-deployment --replicas=3 -n rising
```

- Inspect the changes in the Service:

```
kubectl describe service exposed-deployment -n rising
```

- Clean everything up:

```
kubectl delete deployment exposed-deployment -n rising  
kubectl delete service exposed-deployment -n rising
```

4.6 Service log SV3 - Use multi-port Services

- Create a Pod that exposes two ports:

```
cat <<EOF | kubectl apply -f -  
  apiVersion: v1  
  kind: Pod  
  metadata:  
    name: multi-port-pod  
    namespace: rising  
    labels:  
      app: multi-port-app  
  spec:  
    containers:  
      - name: multi-port-container  
        image: nginx:1.19.6  
        ports:  
          - containerPort: 80  
            name: http  
          - containerPort: 443  
            name: https  
EOF
```

- Expose the Pod through a Service:

```
cat <<EOF | kubectl apply -f -
apiVersion: v1
kind: Service
metadata:
  name: multi-port-service
  namespace: rising
spec:
  selector:
    app: multi-port-app
  ports:
    - name: http
      protocol: TCP
      port: 80
      targetPort: http
    - name: https
      protocol: TCP
      port: 443
      targetPort: https
EOF
```

- Inspect the Service:

```
kubectl describe service multi-port-service -n rising
```

- Clean everything up:

```
kubectl delete pod multi-port-pod -n rising
kubectl delete service multi-port-service -n rising
```

4.7 EndpointSlice log ES1 - Use EndpointSlices to manually group selectorless Services

- Create two Pods and a Service that has no selector:

```

cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: endpoint-pod-1
    namespace: rising
  spec:
    containers:
      - name: endpoint-container
        image: nginx:1.19.6
        ports:
          - containerPort: 80
            name: http
    ---
  apiVersion: v1
  kind: Pod
  metadata:
    name: endpoint-pod-2
    namespace: rising
  spec:
    containers:
      - name: endpoint-container
        image: nginx:1.27.0
        ports:
          - containerPort: 80
            name: http
    ---
  apiVersion: v1
  kind: Service
  metadata:
    name: endpoint-service
    namespace: rising
  spec:
    ports:
      - name: http
        port: 80
        targetPort: 4000
EOF

```

- Inspect EndpointSlices:

```
kubectl get endpointslices -n rising
```

- This Service will not create any EndpointSlice. We need to create one manually:

```
cat <<EOF | kubectl apply -f -
  apiVersion: discovery.k8s.io/v1
  kind: EndpointSlice
  metadata:
    name: endpoint-service-1
    namespace: rising
    labels:
      kubernetes.io/service-name: endpoint-service
  addressType: IPv4
  ports:
    - name: http
      appProtocol: http
      protocol: TCP
      port: 4000
  endpoints:
    - addresses:
      - $(kubectl get pod endpoint-pod-1 -n rising -o jsonpath='{.status.podIP}')
    - addresses:
      - $(kubectl get pod endpoint-pod-2 -n rising -o jsonpath='{.status.podIP}')
EOF
```

- Inspect the EndpointSlice:

```
kubectl describe endpointslice endpoint-service-1 -n rising
```

- Tear everything down:

```
kubectl delete pod endpoint-pod-1 -n rising
kubectl delete pod endpoint-pod-2 -n rising
kubectl delete service endpoint-service -n rising
kubectl delete endpointslice endpoint-service-1 -n rising
```

4.8 Ingress log IG1 - Import an external IngressController, then create an Ingress and expose a Service

- Mark the nodes with ingress-ready:

```
kubectl label node kubeadm-worker1 ingress-ready=true
kubectl label node kubeadm-worker2 ingress-ready=true
```

- Import the NGINX Ingress Controller:

```
kubectl apply -f
↪ https://raw.githubusercontent.com/kubernetes/ingress-nginx/main/deploy/static/provider/kind/deploy.yaml
```

- Wait for the Ingress Controller to be ready:

```
kubectl wait --namespace ingress-nginx --for=condition=ready pod
↪ --selector=app.kubernetes.io/component=controller --timeout=90s
```

- Deploy a Pod and a Service:


```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: ingress-pod
    namespace: rising
  spec:
    containers:
      - name: ingress-container
        image: nginx:1.19.6
        ports:
          - containerPort: 80
            name: http
    ---
  apiVersion: v1
  kind: Service
  metadata:
    name: ingress-service
    namespace: rising
  spec:
    selector:
      app: ingress-app
    ports:
      - name: http
        protocol: TCP
        port: 80
        targetPort: http
EOF
```

- Create the Ingress:

```
cat <<EOF | kubectl apply -f -
  apiVersion: networking.k8s.io/v1
  kind: Ingress
  metadata:
    name: ingress
    namespace: rising
  spec:
    rules:
      - host: ingress.local
        http:
          paths:
            - path: /
              pathType: Prefix
              backend:
                service:
                  name: ingress-service
                  port:
                    number: 80
EOF
```

- Inspect the Ingress:

```
kubectl describe ingress ingress -n rising
```

- List the IngressClasses:

```
kubectl get ingressclasses -n rising
```

- Clean what we created:

```
kubectl delete pod ingress-pod -n rising
kubectl delete service ingress-service -n rising
kubectl delete ingress ingress -n rising
```

- Remove the NGINX Ingress Controller:

```
kubectl delete -f
↪ https://raw.githubusercontent.com/kubernetes/ingress-nginx/main/deploy/static/provider/kind/deploy.yaml
```

- Unlabel the nodes:

```
kubectl label node kubeadm-worker1 ingress-ready-
kubectl label node kubeadm-worker2 ingress-ready-
```

4.9 Proxy Log PX1 - Manually construct a Proxy API call for Nodes, Pods, and Services

- Use `kubectl cluster-info` to obtain endpoint information:

```
cplane=$(kubectl cluster-info | grep 'control plane' | grep -oP 'https://[~/+]' --color=never)
proxy_endpoint=${cplane}
echo $proxy_endpoint
```

- Try the node proxy (the response will be empty, but the call will be successful):

```
first_node=$(kubectl get nodes -o jsonpath='{.items[0].metadata.name}')
./scripts/kube-api.sh $proxy_endpoint api/v1/nodes/${first_node}:10250/proxy
```

- Deploy a Pod and a Service:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: proxy-pod
    namespace: rising
  spec:
    containers:
      - name: nginx
        image: nginx
        ports:
          - containerPort: 80
  ---
  apiVersion: v1
  kind: Service
  metadata:
    name: proxy
    namespace: rising
  spec:
    selector:
      app: proxy
    ports:
      - protocol: TCP
        port: 80
        targetPort: 80
EOF
```

- Try the pod proxy:

```
first_pod=$(kubectl get pods -n rising -o jsonpath='{.items[0].metadata.name}')
./scripts/kube-api.sh $proxy_endpoint api/v1/namespaces/rising/pods/${first_pod}:80/proxy
```

- Try the service proxy:

```
./scripts/kube-api.sh $proxy_endpoint api/v1/namespaces/rising/services/proxy:80/proxy
```

- Clean everything up:

```
kubect1 delete pod proxy-pod -n rising  
kubect1 delete service proxy -n rising
```

4.10 NetworkPolicy log NP1 - Create a NetworkPolicy to restrict traffic to a Pod

- Run some services:

```
kubect1 run web1 --image=nginx --port=80 --labels="app=web" --expose --namespace=default  
kubect1 run web2 --image=nginx --port=80 --labels="app=web" --expose --namespace=rising  
kubect1 run bb --image=busybox --labels="app=bb" --namespace=rising -- sleep 3600
```

A pod is isolated for ingress if there is any NetworkPolicy that both selects the pod and has "Ingress" in its policyTypes. A pod is isolated for egress if there is any NetworkPolicy that both selects the pod and has "Egress" in its policyTypes.

- Create a NetworkPolicy:

```
cat <<EOF | kubectl apply -f -
  apiVersion: networking.k8s.io/v1
  kind: NetworkPolicy
  metadata:
    name: bb-to-web
    namespace: rising
  spec:
    podSelector:
      matchLabels:
        app: web
    policyTypes:
      - Ingress
    ingress:
      - from:
        - podSelector:
            matchLabels:
              app: bb
  ---
  apiVersion: networking.k8s.io/v1
  kind: NetworkPolicy
  metadata:
    name: web-to-web-block
    namespace: rising
  spec:
    podSelector: {}
    policyTypes:
      - Ingress
      - Egress
  ---
  apiVersion: networking.k8s.io/v1
  kind: NetworkPolicy
  metadata:
    name: web-to-web-block
    namespace: default
  spec:
    podSelector: {}
    policyTypes:
      - Ingress
      - Egress
EOF
```

- Assess the NetworkPolicy:

```
kubectl get networkpolicies -A -o wide
kubectl describe networkpolicy bb-to-web -n rising
```

- See if the busybox Pod can reach the web1 Pod:

```
kubectl exec -n rising bb -- wget -q0- http://web1.default | grep Welcome
```

- See if the web1 Pod can connect to the web2 Pod:

```
kubectl exec -n default web1 -- curl -s http://web2.rising
```

- Clean everything up:

```
kubectl delete networkpolicy bb-to-web web-to-web-block -n rising
kubectl delete networkpolicy web-to-web-block -n default
kubectl delete pod web1 -n default
kubectl delete pod web2 bb -n rising
kubectl delete service web1 -n default
kubectl delete service web2 -n rising
```

5 Log setups level 4

5.1 SCHE Storage log ST1 - Install a custom CSI driver and create a StorageClass

- Deploy an NFS server:

```
kubectl create -f  
↳ https://raw.githubusercontent.com/kubernetes-csi/csi-driver-nfs/master/deploy/example/nfs-provisioner/
```

- Install the NFS CSI Driver:

```
ver="v4.8.0"  
repo="https://raw.githubusercontent.com/kubernetes-csi/csi-driver-nfs/$ver/deploy"  
  
kubectl apply -f $repo/rbac-csi-nfs.yaml  
kubectl apply -f $repo/csi-nfs-driverinfo.yaml  
kubectl apply -f $repo/csi-nfs-controller.yaml  
kubectl apply -f $repo/csi-nfs-node.yaml
```

- List and describe the CSINodes:

```
kubectl get csinodes  
kubectl describe csinode kubeadm-master
```

- List and describe the CSIDriver:

```
kubectl get csidrivers  
kubectl describe csidriver nfs.csi.k8s.io
```

- Create a StorageClass: (using nfs-server.default.svc.cluster.local)

```
cat <<EOF | kubectl apply -f -  
  apiVersion: storage.k8s.io/v1  
  kind: StorageClass  
  metadata:  
    name: nfs-storage  
    namespace: rising  
  provisioner: nfs.csi.k8s.io  
  parameters:  
    server: nfs-server.rising.svc.cluster.local  
    path: "/"  
    readOnly: "false"  
  annotations:  
    storageclass.kubernetes.io/is-default-class: "true"  
EOF
```

- List and describe the StorageClasses:

```
kubectl get storageclasses -n rising  
kubectl describe storageclass local-storage -n rising
```

5.2 SCHE Storage log ST2 - Set up a PersistentVolume and a PersistentVolumeClaim with a Deployment

- Create a Deployment with PVCs:

```
cat <<EOF | kubectl apply -f -
  apiVersion: apps/v1
  kind: Deployment
  metadata:
    name: nginx-deployment
    namespace: rising
  spec:
    replicas: 2
    selector:
      matchLabels:
        app: nginx
    template:
      metadata:
        labels:
          app: nginx
      spec:
        containers:
          - name: nginx
            image: nginx:1.19.6
            ports:
              - containerPort: 80
            volumeMounts:
              - mountPath: /usr/share/nginx/html
                name: storage
        volumeClaimTemplates:
          - metadata:
              name: storage
            spec:
              accessModes: [ "ReadWriteOnce" ]
              resources:
                requests:
                  storage: 1Gi
                storageClassName: nfs-storage
EOF
```

- Wait for the Deployment to be ready:

```
kubectl wait --for=condition=Ready deployment/nginx-deployment -n rising
```

- Inspect the Deployment:

```
kubectl get deployments -n rising
```

- Inspect the Pods:

```
kubectl get pods -l app=nginx -n rising
```

- Inspect the PV and PVC:

```
kubectl get pv
kubectl get pvc -n rising
```

- Scale up the Deployment:

```
kubectl scale deployment nginx-deployment --replicas=3 -n rising
```

- Inspect the Pods again:

```
kubectl wait --for=condition=Ready pod -l app=nginx -n rising
kubectl get pods -l app=nginx -n rising
```

- Inspect the PV and PVC again:

```
kubectl get pv
kubectl get pvc -n rising
```

- Scale down to 1:

```
kubectl scale deployment nginx-deployment --replicas=1 -n rising
```

- Inspect the Pods again:

```
kubectl wait --for=condition=Ready pod -l app=nginx -n rising
kubectl get pods -l app=nginx -n rising
```

- Inspect the PV and PVC again:

```
kubectl get pv
kubectl get pvc -n rising
```

- Clean everything up:

```
kubectl delete deployment nginx-deployment -n rising
```

5.3 SCHE Storage log ST3 - Pre-bind a PVC to a PV and use CSI manual provisioning

- Create a PV and a PVC:

```
cat <<EOF | kubectl apply -f -
apiVersion: v1
kind: PersistentVolume
metadata:
  annotations:
    pv.kubernetes.io/provisioned-by: nfs.csi.k8s.io
  name: pv-nginx
spec:
  capacity:
    storage: 10Gi
  accessModes:
    - ReadWriteOnce
  persistentVolumeReclaimPolicy: Delete
  mountOptions:
    - nfsvers=4.1
  csi:
    driver: nfs.csi.k8s.io
    volumeHandle: nfs-server.rising.svc.cluster.local/share##
    volumeAttributes:
      server: nfs-server.rising.svc.cluster.local
      share: /
  ---
kind: PersistentVolumeClaim
apiVersion: v1
metadata:
  name: pvc-nginx
  namespace: rising
spec:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 10Gi
  volumeName: pv-nginx
  storageClassName: ""
EOF
```

- Create a Pod that uses the PV:

```
cat <<EOF | kubectl apply -f -
apiVersion: v1
kind: Pod
metadata:
  name: nginx-nfs-example
  namespace: rising
spec:
  containers:
  - image: nginx
    name: nginx
    ports:
    - containerPort: 80
      protocol: TCP
    volumeMounts:
    - mountPath: /var/www
      name: pvc-nginx
      readOnly: false
  volumes:
  - name: pvc-nginx
    persistentVolumeClaim:
      claimName: pvc-nginx
EOF
```

- Verify that the Pod is running and has the volume mounted:

```
kubectl wait --for=condition=Ready pod/nginx-nfs-example -n rising
kubectl exec nginx-nfs-example -- bash -c "findmnt /var/www -o TARGET,SOURCE,FSTYPE"
```

- Tear everything down:

```
kubectl delete pod nginx-nfs-example -n rising
kubectl delete pvc pvc-nginx -n rising
kubectl delete pv pv-nginx
```

5.4 SCHE Stateful Set log SS1 - Manage a StatefulSet

- Create a StatefulSet with an associated Service:


```

cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Service
  metadata:
    name: nginx
    namespace: rising
    labels:
      app: nginx
  spec:
    ports:
      - port: 80
        name: web
    clusterIP: None
    selector:
      app: nginx
---
  apiVersion: apps/v1
  kind: StatefulSet
  metadata:
    namespace: rising
    name: web
  spec:
    selector:
      matchLabels:
        app: nginx
    serviceName: "nginx"
    replicas: 3
    minReadySeconds: 10 # by default is 0
    template:
      metadata:
        labels:
          app: nginx
      spec:
        terminationGracePeriodSeconds: 10
        containers:
          - name: nginx
            image: registry.k8s.io/nginx-slim:0.24
            ports:
              - containerPort: 80
                name: web
            volumeMounts:
              - name: www
                mountPath: /usr/share/nginx/html
        volumeClaimTemplates:
          - metadata:
              name: www
            spec:
              accessModes: [ "ReadWriteOnce" ]
              resources:
                requests:
                  storage: 200Mi
EOF

```

- Inspect the StatefulSet:

```
kubectl get statefulsets web -n rising
```

- Inspect the Pods:

```
kubectl get pods -l app=nginx -n rising
```

- Scale up the StatefulSet:

```
kubectl scale statefulsets web --replicas=4 -n rising
```

- Inspect the Pods again:

```
kubectl get pods -l app=nginx -n rising
```

- Scale down to 2:

```
kubectl scale statefulsets web --replicas=2 -n rising
```

- Inspect the Pods:

```
kubectl get pods -l app=nginx -n rising
```

- Inspect the PersistentVolumes and the PersistentVolumeClaims:

```
kubectl get pv  
kubectl get pvc -n rising
```

- Clean everything up (StatefulSets retain PVCs by default, so we need to delete them first):

```
kubectl delete statefulset web -n rising  
kubectl delete service nginx -n rising  
kubectl delete pvc www-web-0 www-web-1 www-web-2 www-web-3 -n rising  
kubectl delete pv pv-0  
kubectl delete storageclass local-storage
```

5.5 SCHE Storage log ST4 - Tear down the NFS CSI Driver and its components

- Delete the StorageClass:

```
kubectl delete storageclass nfs-storage
```

- Delete the NFS CSI Driver:

```
ver="v4.8.0"  
repo="https://raw.githubusercontent.com/kubernetes-csi/csi-driver-nfs/$ver/deploy"  
  
kubectl delete -f $repo/csi-nfs-node.yaml  
kubectl delete -f $repo/csi-nfs-controller.yaml  
kubectl delete -f $repo/csi-nfs-driverinfo.yaml  
kubectl delete -f $repo/rbac-csi-nfs.yaml
```

- Delete the NFS server:

```
kubectl delete -f  
↪ https://raw.githubusercontent.com/kubernetes-csi/csi-driver-nfs/master/deploy/example/nfs-provisioner/nfs-provisioner.yaml
```

5.6 SCHE AdmissionControl log AC1 - Create a MutatingWebhookConfiguration and a ValidatingWebhookConfiguration

- Install Gatekeeper:

```
kubectl apply -f  
↪ https://raw.githubusercontent.com/open-policy-agent/gatekeeper/v3.16.3/deploy/gatekeeper.yaml
```

- Wait for the Gatekeeper components to be ready:

```
kubectl wait --for=condition=Ready pod -l app=gatekeeper-controller-manager -n  
↪ gatekeeper-system
```

- List the CRDs created by Gatekeeper:

```
kubectl get crds
```

- Get and list the ValidatingWebhookConfigurations:

```
kubectl get validatingwebhookconfigurations.admissionregistration.k8s.io -n gatekeeper-system
kubectl describe validatingwebhookconfigurations.admissionregistration.k8s.io
↪ gatekeeper-validating-webhook-configuration -n gatekeeper-system
```

- Get and list the MutatingWebhookConfigurations:

```
kubectl get mutatingwebhookconfigurations.admissionregistration.k8s.io -n gatekeeper-system
kubectl describe mutatingwebhookconfigurations.admissionregistration.k8s.io
↪ gatekeeper-mutating-webhook-configuration -n gatekeeper-system
```

- Get and list the PodDisruptionBudgets:

```
kubectl get poddisruptionbudgets.policy -n gatekeeper-system
kubectl describe poddisruptionbudgets.policy gatekeeper-controller-manager -n
↪ gatekeeper-system
```

Creating and managing OPA policies is out of scope for this document.

- Uninstall Gatekeeper:

```
kubectl delete -f
↪ https://raw.githubusercontent.com/open-policy-agent/gatekeeper/v3.16.3/deploy/gatekeeper.yaml
```